

PRACTICAL NO 5 MODULE 2

AIM: Independent Two Sample t-test

OUTPUT:-

The screenshot displays the RStudio interface with the following components:

- Source Panel:** Contains R code for loading a dataset and performing a t-test.

```
R • R 4.5.2 • C:/Users/as993/DATA SET/ > df <- read.csv("professional_hr_employee_dataset.csv")> str(df)'data.frame': 6000 obs. of 10 variables: $ Employee_ID : int 20001 20002 20003 20004 20005 20006 20007 20008 20009 20010 ... $ Department : chr "Marketing" "IT" "Sales" "Sales" ... $ Job_Role : chr "Manager" "Manager" "Intern" "Analyst" ... $ Experience_Years : num 4.5 1.1 10.3 17 6.6 10.5 7.5 4.3 9.7 8.5 ... $ Education_Level : chr "Master" "Diploma" "Master" "Bachelor" ... $ Monthly_Salary : int 169755 60462 128379 35324 105695 60916 71425 141702 98195 170846 ... $ Performance_Score : num 2.2 3.7 3.7 1.3 3.9 4.1 3.5 2 3.7 3.5 ... $ Work_Mode : chr "Office" "Remote" "Remote" "Office" ... $ Overtime_Hours : int 36 14 6 37 29 6 28 12 2 24 ... $ Attrition_Status : chr "No" "No" "No" "No" ...> colnames(df)[1] "Employee_ID" "Department" "Job_Role" "Experience_Years" "Education_Level"> [6] "Monthly_Salary" "Performance_Score" "Work_Mode" "Overtime_Hours" "Attrition_Status"> t_test_two <- t.test(Monthly_Salary ~ Attrition_Status, data = df)> print(t_test_two)
```
- Console Panel:** Shows the output of the t-test.

```
Welch Two Sample t-testdata: Monthly_Salary by Attrition_Statust = 0.5219, df = 1494.5, p-value = 0.6018alternative hypothesis: true difference in means between group No and group Yes is not equal to 095 percent confidence interval:-2353.725 4060.244sample estimates:mean in group No mean in group Yes98144.67 97291.41
```
- Environment Panel:** Lists the objects in the global environment: df (6000 obs. of 10 variables), ecommerce_sales... (5000 obs. of 11 variables), professional_hr... (6000 obs. of 10 variables), t_test_one (List of 10), and t_test_two (List of 10).
- Files Panel:** Shows a file explorer view of the project directory, listing various files including practical notes, processed data, and images.